

EXPERIENCE

Programmer Analyst	American Express	Jan 2021 – Present
- Tools & Tech Stack: Linux, Python, APIs, Spark, SQL, Shell Scripting, Selenium - Responsible for developing and maintaining software solutions for the Credit and Fraud Risk Division. - Interned from Jan to Jun 2021. Full time since Aug 2021.		
iOS Developer, Intern	CS Soft Solutions Pvt. Ltd.	Summer 2019
- Tools & Tech Stack: Xcode, Swift, Stripe API, ARKit, Core Data, SQL, Cocoapods, Git & Gitlab - Food Delivery App: Developed a Food Delivery app based on user story - Augmented Reality Home Planner: Used Apple ARKit to develop an app for a furniture store enabling people to place virtual 3D models of various items in their room to see how it would look and make better decisions.		

SKILLS & TECHNOLOGIES

- C++, Python, Swift, JavaScript, Git for Version Control & Code Collaboration
- Django, ReactJS, NodeJS, SQL, Core Data, Realm, Firestore, PyTorch, fast.ai, Xcode, Unix Based Systems
- Full Stack Web Development, iOS App Development, API Development, Machine Learning, AWS Deployment

PROJECTS

- [Hadoop/Spark] Distributed Computing for Cyber Security:** Practically verified a theoretical technique based on Shannon / General entropy metrics to distribute the network analysis load over a computing cluster to identify and deal with DDOS attacks over WAN using Apache Hadoop & Spark.
- [Brain-Computer Interfacing] Cognitive Neuroscience:** Devised a method to detect muscle artefacts in EEG data using Brain-Computer Interfacing, with a potential for upgraded Brain Mapping by attenuation of the recognized artefacts. It was tested on the Left Frontal Lobe with >95% accuracy.
- [Computer Vision] Accident Detection:** Uses camera feed at traffic intersections to spot road accidents in real time. Secured 3rd place at OctaHacks Hackathon 2019.
- [Software Development] iOS App for Resellers:** Scans product barcodes using iPhone camera, aggregates the price history from different sources using API calls, enabling the reseller to make an informed decision.
- Dark Matter Discovery:** Built a pipeline to visualize and confirm presence of Dark Matter in Cosmic Data based on the principle of Strong Gravitational Lensing, using PyAutoLens.
- Cosmic Artefact Filter:** Trained a deep learning model to flag cosmic ray artefacts in satellite imagery for deeper analysis. Used image data from NASA's MESSENGER Spacecraft orbiting the planet Mercury.
- Quantum Machine Learning:** Researched and demonstrated use of Quantum Machine Learning to distinguish between signal and background events in High Energy Physics.

EDUCATION

Chandigarh	University Institute of Engineering and Technology, Panjab University	Jul 2017 – Jun 2021
- B.E. in Information Technology. CGPA: 8.2/10 (with Honours). Verification Link - Coursework: Data Structures & Algorithms, Database Design, Calculus IV, Linear Algebra, Operating Systems, Computer Architecture, Theory of Computation & Compiler Design, Computer Networking		

ACHIEVEMENTS

SAT Score: 1390/1600 in May 2016 – 800/800 in Mathematics, 590/800 in English, 97th Percentile

6-month Deep Learning Nanodegree	Udacity / Facebook	Nov 2018 – May 2019
- Credential ID: C9HLTFDN. Used PyTorch primarily. - Full Scholarship by Facebook (acceptance rate of 3%) - out of 10,000 global applicants of diverse qualifications. - Coursework: Computer Vision (CNN), Sentiment Analysis (RNN), Generative Adversarial Networks, Deployment.		

VOLUNTEER & ADDITIONAL EXPERIENCE

- Speaker & Mentor:** Moderated Programming Club, UIET. Taught and guided junior CS students
- Udacity Mentor:** Facilitated smooth flow of Deep Learning Nanodegree program on Udacity's behalf